Agentic Workflow Design and n8n Demo

Submission Type	Individual submission
Output Required	Video demo(Google drive URL) + workflow summary(Github readMe) + contribution note if part of a group project
Tool	n8n workflow, using agentic practices and concepts taught in class
Expected Complexity	Meaningful multi-step workflow; group projects should include multiple agents or clearly separated agentic roles
Evaluation Criteria	● Problem-to-workflow mapping ● workflow logic and structure ● AI vs deterministic steps ● practical usefulness, explanation and understanding

1. Assignment Brief

Your task is to identify a real-world problem, frame it clearly, break it down into a meaningful workflow, and implement a working n8n(or similar low code workflow tool) demo that uses agentic practices.

The goal is not to create a simple chatbot. The goal is to demonstrate that you can think like an AI product builder:

- Define the problem
- Design the workflow
- Decide where AI should be used
- Keep deterministic control where needed
- Explain the system clearly

Your final submission must include a Loom video walkthrough showing your workflow demo and explaining the design decisions behind it.

2. Learning Objectives

- Frame a real-world problem into a structured, AI-solvable workflow.
- Decompose a problem into meaningful steps, branches, tools, and outputs.
- Apply agentic design concepts such as role definition, task decomposition, tool use, routing, human approval, and structured outputs.
- Distinguish between AI reasoning steps and deterministic control logic.
- Communicate the workflow clearly through a short technical demo.

3. What You Need to Build

Task Component	Expectation
Select a problem	Choose a practical problem where an AI workflow can create real value. It can be from education, campus operations, placement, productivity, customer support, research, finance, HR, student services, startup operations, or any realistic domain.
Frame the problem statement	Write a clear problem statement: who is the user, what pain point are you solving, why does it matter, and what output should the workflow produce?
Break it into workflow steps	Define the input, processing steps, AI reasoning steps, deterministic rules, decision branches, tools/integrations, and final output.
Build in n8n	Create a working n8n workflow. The workflow should have multiple meaningful nodes and should not be only a single AI prompt call.
Use agentic practices	Use concepts from class such as agent roles, tool use, routing, structured outputs, human-in-the-loop review, validation, branching, and fallback/error handling where relevant.
Record a Loom demo	Record a short demo explaining the problem, workflow, key nodes, where AI is used, where deterministic logic is used, and how the output is useful.

4. Individual Submission Requirement

This is an individual submission. Every student must submit their own Loom video and explanation, even if the underlying workflow was developed as part of a group project.

If you are using a group project, your submission must clearly explain your individual contribution. You should specifically mention:

- which part of the workflow you designed or built
- which nodes, agents, prompts, integrations, or branches you contributed to
- how your part connects with the overall system
- what decisions you made and why

For group projects, the problem should be complex enough to justify multiple contributors. A strong group workflow should ideally include multiple agentic roles, clear decomposition, multiple branches, or meaningful integrations.

5. Suggested Problem Areas

Area	Example Ideas
Student / Campus	Assignment helper with teacher approval, event query router, campus support ticket triage, lost-and-found assistant.
Placement / Career	JD-resume fit analyzer, mock interview workflow, portfolio review workflow, job application tracker.
Startup / Product	Startup idea validator, customer feedback summarizer, MVP feature prioritizer, founder task planner.
Research / Learning	Research paper summarizer with verification, learning roadmap generator, resource recommendation workflow.
Operations / Business	Lead triage, invoice/document review, customer support escalation, meeting summary and action tracker.
Personal Productivity	Email/task prioritizer, study planner, goal tracker, document-to-action workflow.

You may choose a different problem as long as it is practical, clear, and suitable for a workflow-based AI system.

6. Agentic Practices to Demonstrate

- Clear role or responsibility for each AI step, for example extractor, classifier, reviewer, planner, or recommender.
- Structured inputs and outputs, preferably JSON-like or schema-based outputs where possible.
- Tool or integration usage, such as forms, Google Sheets, email, database, HTTP request, document parser, or external API.
- Branching/routing logic based on workflow state or AI output.
- Deterministic checks for validation, thresholds, fit categories, approval status, or routing decisions.
- Human-in-the-loop review for high-impact or risky outputs where appropriate.
- Fallback handling for missing input, incomplete output, or failed tool calls.

7. Submission Deliverables

Deliverable	Details
Loom video	5-8 minutes recommended. Show the working n8n workflow, explain the problem, and run through one sample input/output.
Problem statement	A short written description of the user, pain point, workflow goal, and expected output.

Workflow explanation	Briefly explain the major workflow steps, AI nodes, deterministic nodes, branches, and final output.
Contribution note	Required for students using a group project. Clearly state your individual contribution.
Github Repo	Screenshots, exported n8n workflow JSON, sample input/output, and a README

8. Suggested Loom Video Structure

Time	What to Cover
0:00-0:45	Introduce the problem and target user.
0:45-1:30	Explain the problem-to-workflow breakdown.
1:30-3:30	Walk through the n8n workflow nodes and connections.
3:30-5:30	Run the workflow with sample input and show the output.
5:30-7:00	Explain where AI is used, where deterministic logic is used, and why.
7:00-8:00	Mention limitations, improvements, and your contribution if applicable.

9. Grading Criteria

Criterion	What Will Be Judged	Suggested Weight
Problem-to-workflow mapping	Does the student clearly frame the problem and break it into meaningful workflow steps instead of treating it as one prompt?	20%
Workflow logic and structure	Are the nodes, branches, routing, and sequence logically designed, readable, and functional?	20%
Use of AI vs deterministic steps	Is AI used where reasoning, extraction, summarization, classification, or planning is needed, while deterministic logic handles control, validation, routing, and thresholds?	20%
Practical usefulness	Does the workflow solve a realistic problem and produce a usable output?	20%
Explanation and understanding	Can the student clearly explain what the workflow does, why each major	20%

	step exists, and what their contribution was?	